UNIVERSITY OF LIBERAL ARTS BANGLADESH

Department of Media Studies and Journalism

RESEARCH PROPOSAL

Algorithmic News Feeds and Misinformation Susceptibility Among Young Urban News Consumers in Dhaka: A Mixed-Methods Study

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Abstract

Most urban Bangladeshi university students now get most of their news through algorithmically ranked social media feeds. What we don't know, with any quantitative precision, is how that exposure relates to the everyday skill of telling reliable news from misinformation. Most of the empirical work on this question comes from Western contexts; the few Bangladesh-specific studies are qualitative, single-site, or focused on the supply side rather than on individual susceptibility. This study aims to produce the first comparative quantitative dataset on the relationship between algorithmic feed reliance and misinformation discernment among Dhaka university students, alongside a qualitative account of how students themselves describe their credibility-evaluation practices.

The design is pre-registered, cross-sectional, and mixed-methods. We recruit 300–360 students across six universities in Dhaka — three private (ULAB, BRAC, North South University) and three public (Dhaka University, Jahangirnagar University, Bangladesh University of Engineering and Technology) — stratified by faculty within each site. The quantitative phase measures news consumption across three modalities (Facebook feed, TikTok feed, and AI-mediated news exposure operationally defined in section 6.3), three indices of misinformation discernment (the validated MIST-20 as the primary outcome, an adapted shortened AOT scale and a contextually constructed Bangladesh headline task as robustness checks), share-intent ratings on the headline items as a partial behavioural measure, and four socioeconomic controls beyond institution type. The qualitative phase consists of 18 semi-structured interviews (three per site) covering credibility-evaluation practices, awareness of algorithmic curation, and barriers to source verification.

Our working hypothesis is that heavier reliance on algorithmic feeds is associated with lower misinformation discernment, and that this association is moderated by media literacy training and socioeconomic background. The design is correlational; it cannot establish causality, and findings should not be generalised beyond Dhaka university populations. What it can do is produce a comparative dataset, a set of locally adapted instruments, and a qualitative account of student reasoning that other researchers and Bangladeshi media-literacy programmes can build on.

1. Introduction

On 21 August 2024, a photograph of a woman tied to a tree circulated widely on Facebook across South Asia. Major Indian outlets reposted it as evidence of communal violence against a Hindu school principal in Dhaka, naming her as Geetanjali Barua, Principal of Azimpur Government Girls' School and College. The claim was false. The woman in the photograph was not the principal. By the time anyone said so in print, the image had reached millions of users through the algorithmic feed.

I reported on that incident for *The Business Standard*. What I saw, in real time, was a factually incorrect narrative spreading faster than any correction the local fact-checking infrastructure could mount. That observation is what made me want to do this study. It is not, by itself, evidence for a generalisable claim.

It is an empirical puzzle: does the pattern I saw in one news cycle show up systematically among ordinary news consumers, and if so, who is most affected?

The case for asking is at least quantitative. According to DataReportal's *Digital 2026: Bangladesh* report, Bangladesh had 64.0 million social media user identities in October 2025 — 36.3 percent of the total population — with Facebook the dominant platform at approximately 72.5 million accounts (Kemp, 2025). In traditional media environments, editorial gatekeeping filters information before publication, however imperfectly. In platform environments, ranking is done by algorithms designed to maximise engagement. Content that provokes outrage, fear, or moral shock tends to spread further than content that is accurate or corrective. On Twitter at least, Vosoughi, Roy, and Aral (2018) demonstrated that false news spread faster and further than true news, with their controlled analyses attributing the differential primarily to human re-sharing rather than to bots. Whether that pattern generalises to Bangladeshi news consumers using Facebook, TikTok, and increasingly AI-mediated news exposure is what this study asks.

There are reasons to think Bangladesh is a particularly important site for the question. Platform penetration is high, media literacy infrastructure is thin, and the dominant ranking systems are designed, deployed, and governed outside the country by companies with limited direct accountability to Bangladeshi users or norms. Sajadieh et al. (2026), in the Stanford AI Index 2026, report that four in five U.S. high school and college students now use generative AI for school-related tasks, suggesting a trajectory that comparable urban Bangladeshi populations are likely to follow even without good local statistics.

What's missing is empirical work. Peer-reviewed quantitative research examining the relationship between algorithmic feed exposure and measurable misinformation susceptibility in Bangladesh is, on a systematic JSTOR, Scopus, and Google Scholar search (terms: misinformation, Bangladesh, algorithm, social media, conducted January 2026), effectively absent. This study is an attempt at that work: a mixed-methods, two-phase, six-site comparison of private and public university students in Dhaka, with quantitative measurement of platform reliance and misinformation discernment followed by interviews that ask students to describe their own credibility-evaluation practices.

The design is cross-sectional and correlational. It cannot establish causality. What it can do is produce the first comparative dataset of its kind for Bangladeshi university students and a set of locally adapted instruments that other studies can borrow.

2. Research Objective

To examine associations between algorithmically curated social media news consumption and misinformation susceptibility among university students in Dhaka, comparing private and public university samples, and to explore the social and cognitive mechanisms that participants themselves identify as shaping their epistemic behaviour in platform-mediated information environments.

3. Research Questions

Primary research questions

RQ1. Is heavier reliance on algorithmically curated social media news consumption associated with lower misinformation discernment, as measured by the MIST-20, among university students in Dhaka?

RQ2. Do specific consumption patterns — disaggregated by modality (Facebook feed, TikTok feed, AI-mediated news exposure), by frequency, and by self-reported awareness of algorithmic curation — differ in their association with discernment?

RQ3. Do discernment patterns differ between private and public university students, and to what extent do socioeconomic and digital-capital variables account for any observed institutional difference?

RQ4. To what extent do scores on the MIST-20, an adapted shortened AOT scale, and a contextually constructed Bangladesh headline task converge as measures of discernment in this sample?

Qualitative sub-questions

RQ5. How do young news consumers describe their own processes for evaluating the credibility of news content encountered through social media feeds?

RQ6. What barriers to critical news evaluation do participants identify, and what role do they assign to platform design, educational background, and social environment in shaping their epistemic habits?

4. Literature Review

Keywords: *Algorithmic news curation · platform misinformation · media literacy · epistemic vulnerability · digital news consumption · Bangladesh*

4.1 The platform turn in news consumption — and what we now know and don't

Social media platforms have displaced traditional media institutions as the primary news gateway for large segments of the global population (Newman et al., 2025). In Bangladesh, the Digital 2026: Bangladesh report records 64.0 million social media user identities in October 2025, 36.3 percent of the total population, with Facebook the dominant platform at approximately 72.5 million accounts and growth of around 8.5 million users (15.3 percent) year on year (Kemp, 2025).

This shift is not just a change in distribution channel. Editorial decisions about newsworthiness and reliability are made — however imperfectly — by human journalists and editors operating within professional norms. Platform ranking is made by algorithmic systems designed to optimise for user engagement.

Early scholarship described this shift in strong filter-bubble or echo-chamber terms (Pariser, 2011; Sunstein, 2017). The empirical record since has substantially complicated that picture. Bakshy, Messing, and Adamic's (2015) Facebook-internal study found cross-cutting content exposure was reduced more by users' own click choices than by the algorithm. Eady, Nagler, Guess, Zilinsky, and Tucker (2019), using behavioural Twitter data, found echo-chamber exposure was less severe than the popular framing suggested. Most recently, the 2023 Meta-collaboration studies (Guess et al., 2023; Nyhan et al., 2023) found surprisingly small effects of algorithm changes on political attitudes over weeks-long intervention windows. Bagchi et al. (2024), in a *Science* eLetter critiquing the Guess et al. (2023) study, argued that Meta's emergency algorithm changes around the 2020 U.S. election contaminated the Guess et al. baseline, meaning the "standard" algorithm in that study was already less misinformation-promoting than usual. Metzler and Garcia (2024) review the broader evidence and conclude that the role of algorithms in misinformation and polarisation is "far from straightforward" and that substantial further empirical research is needed.

The takeaway for this study is that the literature is more contested than the popular framing suggests. Algorithmic feeds plausibly contribute to misinformation exposure, but individual-level factors (motivated reasoning, partisanship, prior beliefs, analytical thinking) also matter substantially (Pennycook & Rand, 2019; Bail et al., 2018). The relationship between exposure and behaviour is bidirectional and mediated by individual differences. This study is designed to examine the association without presupposing a single causal direction.

4.2 Algorithmic ranking and misinformation propagation

A substantial body of research documents that false or low-quality content propagates differently on platforms than accurate content does. Vosoughi, Roy, and Aral (2018), in their study of Twitter rumour cascades, found that false news spread faster and further than true news — and that controlled analyses attributed the difference primarily to human re-sharing rather than to algorithmic amplification or bots (in their experimental conditions, bots accelerated true and false content at roughly equal rates). Brady, Wills, Jost, Tucker, and Van Bavel (2017) demonstrated, also on Twitter, that emotionally arousing and morally provocative content receives systematic amplification through human-driven sharing.

The mechanism on engagement-ranked feeds therefore appears to be a feedback loop: human users react more strongly to certain kinds of content; the algorithm picks up on the engagement signal and surfaces similar content more often. Whether the algorithm's contribution to that loop is "small but consequential" or "the main driver" is contested. Guess et al. (2023) found that restructuring Facebook's feed away from engagement-based ranking modestly reduced exposure to low-quality content but did not move users' political attitudes within the intervention window. Bagchi et al. (2024) argued the Guess et al. comparison was confounded by Meta's emergency election-period changes to the standard algorithm.

In the South Asian context, evidence on the role of algorithmic feeds in misinformation propagation is thinner but suggestive. Amnesty International's (2022) investigation of Meta's role in the spread of anti-Rohingya content in Myanmar documented engagement-ranking dynamics in a context of weak local regulatory and media literacy infrastructure. Rahaman, Efaz, Rajon, Ashraf, and Juearia (2026), examining Bangladesh's July 2024 uprising through 112 verified fact-checking reports, 87 news articles, and 15 interviews with journalists and fact-checkers, document a surge in politically charged disinformation amplified through social media networks that outpaced local fact-checking capacity. Their evidence does not isolate the algorithmic from the human-network contribution but supplies a strong contextual case that platform-mediated misinformation is consequential in Bangladesh.

4.3 Media literacy and individual susceptibility

Media literacy — the capacity to access, analyse, evaluate, and create media in various forms — has long been identified as a critical competency for democratic citizenship (Aufderheide, 1993). In the digital context, it spans what scholars variously term news literacy, digital literacy, and critical information literacy (Mihailidis & Viotty, 2017).

The cross-individual variance in misinformation susceptibility is substantial and only partly explained by general education or intelligence. Pennycook and Rand (2019) argue that susceptibility to partisan misinformation is better explained by analytical thinking — willingness to deliberate rather than rely on intuitive judgments — than by motivated reasoning. Roozenbeek, Schneider, Dryhurst, et al. (2020), in a 26-country study during the COVID-19 pandemic, found analytical thinking and trust in scientists predicted lower susceptibility. The relevance for Bangladesh is direct: university-educated youth here have high general academic attainment but often minimal explicit training in source evaluation or claim verification.

Roozenbeek et al. (2022) further argue that misinformation discernment is reasonably stable across question framings and that myside bias and partisanship explain more of the variance than analytical-thinking measures do. The picture that emerges is one of multiple, partially overlapping individual-difference predictors: analytical disposition, prior media literacy, partisan motivation, and platform-use patterns.

The concept of *epistemic vulnerability* — structural susceptibility to false belief produced by the design of one's information environment rather than by individual cognitive failure — provides a useful analytical frame (Fallis, 2021). But, as Bail et al. (2018) demonstrate, individual agency is not absent: users choose which platforms to use, how frequently, and for what purposes. Epistemic vulnerability is best understood as the interaction of platform architecture and individual characteristics, not as a structural determinant.

4.4 Digital news consumption in Bangladesh: the research gap

Empirical research on digital news consumption in Bangladesh is limited relative to the scale of the phenomenon. Muzykant, Hossain, Muqsith, and Fatima (2022) document structural deficits in media literacy that leave Bangladeshi citizens, particularly urban youth, vulnerable to platform-mediated misinformation. Haque, Yousuf, Alam, Saha, Ahmed, and Hassan (2020) document persistent institutional constraints on Bangladesh's fact-checking ecosystem, including resource limitations and political pressure. Rahaman et al. (2026) document the surge in algorithmically amplified disinformation during the July 2024 uprising. Each of these is informative but supply-side or qualitative; none provides a quantitative measure of individual susceptibility paired with consumption-pattern data.

The algorithmic systems that mediate Bangladeshi news consumption are designed, deployed, and governed almost entirely outside the country. Empirical research on how those externally governed systems interact with the epistemic behaviour of Bangladeshi news consumers is a prerequisite for any serious national policy response. Peer-reviewed quantitative work examining the relationship between algorithmic feed exposure and measurable misinformation discernment in a Bangladeshi sample is, on a systematic JSTOR, Scopus, and Google Scholar search (January 2026), effectively absent. This study is designed to address that gap.

5. Conceptual Framework

The framework integrates platform studies, media literacy research, and the sociology of knowledge to describe the landscape of factors this study examines. It is not a linear causal pipeline. The data is cross-sectional and the relationships are best understood as a set of associations operating across three analytic levels.

Figure 1: Conceptual Framework

Algorithmic Epistemic Vulnerability

Three interlocking analytic levels through which vulnerability is examined

Level 1 — Structural · Platform architecture shapes the information environment

Engagement optimisation	**Emotional amplification**	**Accuracy deprioritised**
Clicks, shares, time-on-platform	Outrage and fear rewarded	Epistemic quality not optimised for

↓ *enables* ↓

Level 2 — Behavioural · Habitual consumption displaces critical evaluation

Passive scrolling	**Source bypass**	**Fluency bias**
Mobile-first, feed-dependent	No direct outlet engagement	Familiar = credible heuristic

↓ *associated with* ↓

Level 3 — Cognitive (current state) · Distribution of discernment in the study population

Reduced verification	**Lower source literacy**	**Discernment variance**

Claim-checking habits weaker	Provenance evaluation weaker	Distribution measured cross-sectionally
Levels are mutually reinforcing, not strictly causal. Measured by: Quantitative survey (Levels 1–2) + Semi-structured interviews (Level 3). Moderators tested: media literacy, partisanship, analytical thinking, socioeconomic background.		

Source: Researcher's own framework, adapted from Pennycook & Rand (2019), Metzler & Garcia (2024), and Fallis (2021).

Level 1 — Structural. Platform architecture shapes the information environment. Engagement-ranked feeds prioritise emotionally arousing content over epistemically reliable content. This is a structural tendency well documented in the literature, not an assumption of this study.

Level 2 — Behavioural. Reliance on algorithmically curated feeds as a primary news source is associated, in existing research, with reduced practices of critical news evaluation — direct engagement with original sources, cross-referencing, and deliberate verification. Individual differences in motivation, prior beliefs, and analytical disposition moderate this association.

Level 3 — Cognitive (current state). The study examines current patterns of misinformation discernment — not degradation over time, which a cross-sectional design cannot measure, but the distribution of discernment scores, source-evaluation practices, and susceptibility to engagement-cued content as they exist in the study population at the time of measurement.

The framework treats motivated reasoning, partisanship, prior media literacy training, peer network effects, and socioeconomic factors as substantive moderators that may explain variance in discernment independently of platform reliance. The quantitative phase tests the relative predictive contribution of these variables. The framework is the researcher's own integrative reading of Pennycook & Rand (2019), Metzler & Garcia (2024), and Fallis (2021); it is offered as a way of organising the analysis, not as a tested model.

6. Methodology

6.1 Research design

This study uses a sequential explanatory mixed-methods design (Creswell & Plano Clark, 2023). The quantitative phase establishes the patterns and correlates of misinformation discernment across the sample. The qualitative phase explores how participants make sense of those patterns in their own terms. The design is cross-sectional; no causal inferences are drawn.

The study will be pre-registered on the Open Science Framework before data collection. The pre-registration will specify the primary outcome (MIST-20 score), the robustness measures (adapted AOT scale and Bangladesh headline task), the convergence-handling rule (see section 6.3), the analysis plan including all covariates and moderators, the headline-task exclusion rule, and the contingency plan for qualitative sampling under both confirmed and disconfirmed quantitative findings (see section 6.5).

6.2 Study setting and target population

The study uses a purposive multi-site design across six universities in Dhaka, three private and three public. The selection logic is theoretically motivated rather than convenience-based: digital-capital variables (home internet quality, primary device, schooling medium) and broader socioeconomic variables vary systematically across institution type. Including both maximises variation on the moderator variables this study examines.

The six sites and the selection rationale:

Private universities:

1. **University of Liberal Arts Bangladesh (ULAB)** — established mid-tier private, English-medium, central Dhaka. The lead researcher has direct institutional access.
2. **BRAC University** — higher-tier private with strong research infrastructure; broad disciplinary mix.
3. **North South University (NSU)** — large-enrolment private with a mixed socioeconomic intake spanning the upper-middle to lower-middle bracket of the private sector.

Public universities:

4. **University of Dhaka (DU)** — elite national public university with the largest and most diverse undergraduate population.
5. **Jahangirnagar University (JU)** — residential public university with a distinct campus culture and mixed socioeconomic intake.
6. **Bangladesh University of Engineering and Technology (BUET)** — elite technical public with predominantly Bangla-medium intake and a different disciplinary profile from DU and JU.

Institution type is entered as a binary predictor (private/public) in the regression. Site is entered as a fixed effect to absorb between-site variance that institution type does not explain. The institution-type effect is interpretable only after controlling for individual-level socioeconomic variables (see 6.3); the proposal does not claim that any institution-type effect reflects institutional culture rather than student composition until that test has been run.

The target population at each site is undergraduate students aged 18–28 who use Facebook, TikTok, or AI-mediated news sources at least once per week. No restriction is placed on faculty, year of study, or gender.

6.3 Phase 1: Quantitative survey

Sample, recruitment, and power

Target sample: 300–360 students across the six sites, with 50–60 per site. Recruitment is run by a research assistant outside the lead researcher's teaching reach. Stage one stratifies by institution type (private/public) and then by site. Stage two stratifies within each site by faculty to ensure disciplinary variation. The sampling frame at each site is the registrar's enrolment roster, accessed under a data-sharing agreement negotiated with each institution before data collection. Where institutional access is denied at a site, that site is replaced and the limitation noted explicitly.

Participants receive a 500 BDT mobile recharge or equivalent on completion. Compensation is disclosed in the recruitment materials and in the consent form.

Power: at $f^2 = 0.05$ (a more empirically defensible "small" effect in media-effects regressions) with 12 predictors at $\alpha = .05$ and power = .80, the required N is approximately 300, matching the recruited target. The conventional medium-effect assumption ($f^2 = 0.15$) yields a minimum N of 118; the higher target provides headroom for missing data, attrition, and subgroup analyses by institution type. Site-level inference is acknowledged as limited: a between-site contrast at $N \approx 50$ vs $N \approx 50$ detects Cohen's $d \geq .56$, which is moderate-to-large; smaller site-level effects cannot be reliably detected and are not claimed.

Survey instrument

The survey has four sections.

Section A — demographics and socioeconomic background. Age, gender, faculty, year of study, prior media literacy education, and four socioeconomic / digital-capital variables: (i) home internet access quality on a 4-point scale, (ii) primary device used for news consumption, (iii) schooling medium (Bangla / English / mixed), and (iv) parental education level (highest of two parents), used as a proxy for household socioeconomic status. These are entered as controls in the regression to test whether observed institution-type effects survive socioeconomic adjustment.

Section B — news consumption patterns. Frequency-of-use and proportion-of-news items are collected separately for three modalities:

- *Facebook algorithmic feed.* Defined as news encountered through scrolling the main Facebook feed, including posts from pages, friends, and groups, with no direct visit to a news outlet.
- *TikTok feed.* Defined as news encountered through the TikTok For You page, including news-style short videos from creators or news outlets.
- *AI-mediated news exposure.* Operationally defined as news content obtained from (a) explicit queries to ChatGPT, Gemini, Claude, or comparable assistants; (b) AI Overview blocks at the top of Google search results; (c) lock-screen or in-app AI summaries (Pixel, iOS, news apps). Each sub-component is captured separately and a composite index constructed only after exploratory analysis confirms the sub-components covary.

Additional items measure direct news consumption (visiting a news outlet's site or app), self-reported awareness of algorithmic curation, and political-news interest.

Section C — misinformation discernment (the primary outcome and two robustness checks).

The primary outcome is the **MIST-20** (Maertens, Götz, Schneider, Roozenbeek, et al., 2024), the 20-item validated misinformation susceptibility test, adapted into Bengali using the three-stage procedure described below. The MIST-20 yields three scores: real-news detection, fake-news detection, and overall veracity discernment; the overall discernment score is the primary outcome.

Two robustness measures accompany the MIST-20:

- An **adapted shortened AOT scale** (seven items, drawn from the Haran, Ritov, and Mellers, 2013 shortened version of Stanovich and West's Actively Open-Minded Thinking scale), measuring disposition toward evidence-based reasoning.
- A **Bangladesh headline discernment task** of 20 items (10 accurate, 10 fabricated), constructed for ecological validity in the local context. Fabricated items are drawn from genuine misinformation events documented in Bangladeshi fact-checking records (Rumor Scanner, Fact Watch), excluding the Azimpur incident and any incident the lead researcher has previously reported on (full exclusion list documented in the pre-registration). Each headline is accompanied by a 5-point share-intent item ("How likely would you be to share this with a friend or family member?"), which provides a partial behavioural supplement to the stated-judgment task.

Convergence-handling rule (pre-registered): the MIST-20 overall discernment score is the primary outcome of all hypothesis tests. The adapted AOT scale and the Bangladesh headline task are reported as robustness checks. If the three measures converge (pairwise $r \geq .40$), a composite z-score average is reported as a supplementary outcome. If they diverge (pairwise $r < .40$), each is analysed separately and the divergence is discussed substantively, not treated as a measurement failure.

Section D — media literacy and AI literacy. Self-reported prior media literacy training; familiarity with how AI language models generate text; understanding of their limitations.

Scale adaptation procedure

The MIST-20, the AOT short scale, and the Bangladesh headline task are adapted using a four-stage procedure. First, items are reviewed by the researcher and supervisor for conceptual equivalence; items with no equivalent local referent are replaced with contextually equivalent items, with the substitution rules documented. Second, instruments are forward-translated into Bengali by an independent bilingual translator (not the lead researcher), back-translated into English by a second independent bilingual translator, with discrepancies reconciled by both translators in consultation with the supervisor. Third, adapted instruments are pilot-tested with 60 students (not 20, to support stable item-level decisions); items where accuracy clusters above 90 percent correct or below 30 percent correct are flagged for replacement. Fourth, internal consistency (Cronbach's α) is calculated on the pilot data; items reducing α

materially are revised. The full adaptation log is reported in the methods section of the final paper and posted with the pre-registration.

Analysis

Descriptive statistics characterise the sample's consumption patterns and discernment scores, reported overall and disaggregated by institution type. Multiple linear regression identifies predictors of MIST-20 discernment while controlling for demographics, the four socioeconomic / digital-capital variables, and site (fixed effect). Institution type is entered as a binary predictor; the institution-type coefficient is reported with and without socioeconomic adjustment to make the contribution of student composition versus institutional context visible. Effect sizes (Cohen's f^2) accompany significance values. Where institution type emerges as a significant predictor net of socioeconomic controls, post-hoc analysis explores which specific institutional characteristics may mediate the effect. All inferences are reported as correlational; no causal claims are drawn from the regression.

6.4 Phase 2: Qualitative interviews

Sample and recruitment

After the quantitative phase, 18 participants are recruited for semi-structured interviews using purposive sampling, three per site. Within each site, participants are selected for variation across MIST-20 discernment level (high, medium, low) and platform reliance pattern. Outlier participants — those with high platform reliance but high discernment — are specifically prioritised across sites to surface potential protective factors.

A pre-registered contingency: if the quantitative phase does not yield a significant association between platform reliance and discernment, qualitative recruitment shifts to maximum-variation sampling across discernment levels, with the interview guide adjusted to probe how high-discernment students reason about platform content regardless of consumption volume.

Interview guide

Interviews are semi-structured, approximately 45–60 minutes, default English with Bengali permitted for sensitive material (ULAB and most private sites operate primarily in English; DU/JU/BUET participants may prefer Bengali). The guide covers five thematic areas: news consumption habits and platform experience; awareness of algorithmic curation; credibility-evaluation practices, including walkthroughs of recent examples; misinformation experience, including instances of successful identification; and perceived barriers and enablers of critical news evaluation.

Transcription, translation, and coding

Audio is recorded with informed consent. Transcription is done by an independent research assistant (not the lead researcher) verbatim in the language of the interview. Bengali transcripts are forward-translated by an independent bilingual translator; the lead researcher reconciles translation choices in consultation with the translator and a bilingual second coder. Critically, the bilingual second coder works from the Bengali source transcript, not from the English translation, to detect any interpretive drift introduced by translation. A 20 percent double-coded subset checks coding reliability (target Cohen's $\kappa \geq .70$).

Thematic analysis follows Braun and Clarke's (2006) six-phase framework. Analysis is guided by but not limited to the conceptual framework, allowing for emergent themes. Original Bengali excerpts are retained in the data record alongside translations. Member checks on translated quotes are offered to all participants who use Bengali in the interview.

6.5 Integration of quantitative and qualitative data

Quantitative and qualitative findings are integrated at the interpretation stage using a connecting strategy: the quantitative results identify what patterns exist; the qualitative findings explore how participants make sense of those patterns. Two pre-specified integration scenarios:

Scenario A (predicted association confirmed): qualitative sampling targets outliers (high reliance, high discernment) and interviews probe what protective factors these participants describe.

Scenario B (predicted association weak or absent): qualitative sampling shifts to maximum variation on discernment, and interviews probe the reasoning practices of high-discernment students irrespective of consumption volume.

Where quantitative and qualitative findings diverge in either scenario, the divergence is treated as analytically significant, not as a failure of one method.

6.6 Reflexivity and structural mitigations

The lead researcher's prior professional experience as a journalist who reported on the Azimpur misinformation incident creates an analytical asset and a potential bias. The asset is contextual knowledge and network access. The risk is that the research design and analysis are unconsciously shaped toward confirming prior observations.

The reflexivity is addressed through writing — a positionality statement, a reflective journal — and through structural separation of roles:

1. Recruitment is run by a research assistant, not by the lead researcher.
2. Survey data collection is administered through an online platform with participants accessing it independently; the lead researcher does not handle paper-based or in-person submission.
3. Interview transcription is done by an independent research assistant.
4. Forward translation of Bengali transcripts is done by an independent bilingual translator.
5. The bilingual second coder of qualitative data works from the Bengali source, not from the English translation.

6. A subset of quantitative analyses (the primary regression model) is replicated by the supervisor independently before findings are reported.
7. The Azimpur incident and any incident the lead researcher has previously reported on are excluded from the headline task by pre-registered rule.

These separations make the researcher's interpretive choices visible at every stage of the pipeline and reduce the surface area for unconscious confirmation bias. They do not eliminate the risk, and the final paper carries an explicit reflexivity statement disclosing the researcher's professional background and these mitigations.

6.7 Ethical considerations

All participants receive a plain-language information sheet explaining the study's purpose, procedures, and their right to withdraw at any time without consequence. Written informed consent is obtained before survey completion and verbal recorded consent before interviews. Survey data is stored anonymously; interview participants are assigned pseudonyms. Data is stored on encrypted institutional storage and accessed only by the research team. Ethical approval is sought from ULAB's research ethics committee. For the five partner institutions, the ULAB approval documentation is shared with relevant departmental contacts; where a partner institution requires separate clearance, that process is initiated in parallel during Weeks 2–4 of the workplan.

6.8 Limitations

This study has seven limitations that should be acknowledged explicitly.

Cross-sectional design. The design permits correlation analysis only; it cannot establish causal direction or change over time.

Sample bounds. The study is bounded to urban Dhaka university students. Findings should not be generalised to rural Bangladesh, non-university populations, or university students outside Dhaka without further research.

Multi-site coordination risk. Six-site execution introduces delays in institutional access, variation in site coordinator engagement, and the possibility that access is denied at one or more sites. The design includes a one-week preparation buffer (see workplan) and a site-replacement contingency; the risk cannot be eliminated.

Self-report and stated-judgment. Self-reported news consumption is subject to social desirability and recall limitations. The headline task is a stated-judgment measure; share-intent items provide a partial behavioural supplement but ecological validity remains lower than for actual platform behaviour.

New / adapted instruments. The Bangladesh headline task is newly constructed; the MIST-20 and AOT short scale are adapted into Bengali. The N = 60 pilot supports basic item functioning checks but is not sufficient for full psychometric validation. The reliability of these instruments in Bangladeshi samples is reported transparently and treated as a contribution to the methodological literature rather than as a settled measurement.

Researcher prior involvement. The lead researcher previously reported on a misinformation incident relevant to the study. The structural mitigations in 6.6 are designed to manage but not eliminate this risk.

Site-level inference is limited. N ≈ 50 per site supports descriptive characterisation and contrasts of moderate-to-large effects (d ≥ .56); finer-grained between-site comparisons are not powered.

7. Workplan

The 22-week timeline runs from May to October 2026. The two-week buffer relative to the prior plan absorbs preparation-phase coordination risk (institutional access, site coordinator confirmation, instrument piloting). The workplan has four phases: Preparation (Weeks 1–6), Data Collection (Weeks 7–14), Analysis (Weeks 13–18), and Write-Up (Weeks 18–22), with intentional overlap between adjacent phases.

#	Activity	Wk 1–4	5–6	7–8	9–10	11–12	13–14	15–16	17–18	19–20	21–22
1	Finalise instruments and interview guide	✓									
2	Ethics approval (ULAB + partner sites)	✓	✓								
3	Institutional access letters (5 partner sites)	✓	✓								
4	Site coordinator confirmation		✓								
5	Independent forward/back translation of MIST-20, AOT, headline task	✓									
6	Pilot survey (N = 60, ULAB)		✓								
7	Refine instruments based on pilot		✓								
8	Survey administration — three private sites			✓	✓						
9	Survey administration — three public sites				✓	✓					
10	Quantitative cleaning and analysis					✓	✓	✓			
11	Purposive interview recruitment				✓	✓					
12	Semi-structured interviews (third-party interviewer)					✓	✓	✓			
13	Independent transcription + translation						✓	✓	✓		
14	Qualitative thematic coding							✓	✓	✓	

#	Activity	Wk 1–4	5–6	7–8	9–10	11–12	13–14	15–16	17–18	19–20	21–22
15	Quant–qual integration								✓	✓	
16	Write-up and discussion								✓	✓	✓
17	Supervisor review and revision									✓	✓
18	Final draft and submission										✓

8. Expected Contributions

The study aims to make four contributions, with scope appropriate to a single-city, cross-sectional, mixed-methods design.

First, it produces the first comparative quantitative dataset on algorithmic news consumption and misinformation discernment across private and public university students in Dhaka — a sample frame underrepresented in the South Asian digital media literature and one that prior single-institution studies cannot address.

Second, by disaggregating consumption across Facebook feed, TikTok feed, and AI-mediated news exposure, and by controlling for four socioeconomic and digital-capital variables, the study produces more precise estimates of which specific consumption patterns are associated with discernment than prior single-platform studies have achieved.

Third, the institution-type contrast, run with and without socioeconomic adjustment, directly tests whether observed institutional differences in discernment reflect student composition or something specific to the institutional context — a question with direct implications for where media-literacy interventions should be targeted.

Fourth, the study contributes a set of locally adapted instruments (Bengali MIST-20, adapted shortened AOT, Bangladesh headline task) and a methodological template (independent translation, role-separated reflexivity, MIST-20 as primary outcome with two robustness checks) that other Bangladeshi institutions can adapt or contest.

The study is positioned as foundational empirical work for a future research agenda this study cannot itself pursue: longitudinal tracking of consumption–discernment associations, nationally representative sampling across Bangladesh's tertiary system, and experimental manipulation of feed exposure. The present study establishes the baseline comparative dataset and the locally validated instruments that such future work requires.

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